Cover Letter

Dear Dr. Rus and Dr. Gaban, Guest Editors,

We would like to submit the enclosed manuscript entitled "Siting Urban Green Infrastructure by Workplace Rather than Residence: Heat-Attributable Work Capacity Loss Among Young Workers under Exposure–Response Uncertainty" for consideration in Sustainability, for the Special Issue "Thermal Mitigation Effects of Green and Blue Infrastructure and Urban Sustainability from an Interdisciplinary Perspective". The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

The Special Issue asks how the strategic implementation of green and blue infrastructure mitigates urban heat, and it lists the socioeconomic impacts of heatwaves, heat mitigation strategies, urban greenery, urban form and urban ventilation among its themes. The enclosed manuscript takes up a question that sits inside that list but has not been asked quantitatively: when a city decides where to plant, whose exposure should it count? Green infrastructure is sited almost everywhere by residential population and residential deprivation, while heat damages work where work happens — and entry-level outdoor employment, through which young Europeans enter the labour market, is concentrated in precisely the industrial and logistics districts where few people live.

The main findings are as follows.

(1) Siting rule dominates. Weighting a fixed planting budget by exposed workplaces rather than by residents protects 3.05 times as many hours of young workers' work capacity, against a control rule that targets the same heat but counts residents. Following residential population is worse than not targeting at all: the uniform rule outperforms it.

(2) The advantage is conditional, and the condition is measurable. It is absent where workplaces and homes coincide and reaches 5.6-fold where they are opposed, crossing into materiality once the correlation between workplace and residential density falls below roughly 0.25. Cities can compute that correlation from data they already hold, so the recommendation is checkable rather than universal.

(3) A negative finding we think the field needs. The magnitude of the benefit is not identified. Across 4000 draws over 28 parameters, five published exposure–response functions, four climate settings and the geography of employment, the estimated loss varies by more than an order of magnitude and vanishes entirely in 6.7 per cent of draws, because the published response functions disagree about whether European summer conditions cause any loss at all. Point estimates in this genre are reporting a choice of function. Rankings survive; levels do not.

(4) Transparency about what the study is. This is a model-based assessment, and the manuscript says so in the abstract, the methods and the data availability statement. The city is synthetic, because the question — where cooling ought to go, as against where it goes — is counterfactual and no observational design answers it. Its residential density and its four climate settings are anchored to the 556 EU-27 urban centres of the Global Human Settlement Layer and to Berkeley Earth country summer temperatures. All 28 parameters are listed in an appendix with their provenance: 8 are taken from cited sources and 20 are modelling choices, which are swept across declared intervals rather than defended.

We believe the manuscript fits the Special Issue closely. It bridges microclimate, urban design and labour economics, which is the cross-sectoral perspective the call champions, and it delivers an actionable insight: the 3–30–300 rule already names workplaces alongside homes and schools, but compliance has only ever been evaluated for dwellings, and our results indicate that the two definitions of the population served are not interchangeable. We hope the manuscript proves suitable for the Special Issue, and we would be pleased to revise it in the light of the reviewers' comments.

Yours sincerely,

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